Timothy Hsu

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EDUCATION

Rutgers University Honors College – New Brunswick, New Jersey **Expected Graduation, May 2028**
B.S. in Computer Science & Mathematics GPA 3.94
Relevant Coursework: Data Structures and Algorithms, Discrete Structures, AI in Math, Probability and Statistics
Affiliations**:** Rutgers Chinese Student Organization, Break Through Tech, Quantitative Finance Club

EXPERIENCE

TRACE AI Lab **New Brunswick, New Jersey**
Machine Learning Researcher *May 2026 – Present*

- Built an ML interpretability pipeline in Python (*GLIMPSE: Geometric Lens for Inspecting Multiplicity of Sets and Explanations*) over **5,000+ records**, benchmarking **4 dimensionality reduction algorithms**, reducing projection selection to a single automated run.
- Engineered **3 geometric fidelity metrics** (Pearson/Spearman distance correlation, trustworthiness), quantifying projection distortion, enabling automated evaluation across high-dimensional feature spaces.
- Designed modular codebase across 4 core libraries with standardized interfaces, cutting dataset and algorithm onboarding time **~80%,** enabling rapid experimentation across **10+ configurations.**
- Developing a novel autoencoder-based projection with custom multi-term loss functions, preserving counterfactual distances and uncertainty geometry under parameter uncertainty.

Eleanna Kara Neurodegenerative Disease Lab **Piscataway, New Jersey**
Lead Programmer & Research Assistant *Sept 2025 – Present*

- Developed Python-based data processing pipelines to analyze **800+ microscopy images,** removing analysis bias by **>50%.**
- Designed and optimized segmentation, thresholding, and contour-detection algorithms to extract **15–20 quantitative cell features** (e.g., soma area, circularity, neurite length), increasing analysis throughput by **5x.**
- Automated QC, visualization, and summary-statistics generation, enabling rapid iteration across **8+ experimental batches** and improving reproducibility of downstream neurobiological analyses.
- Modularized codebase and implemented reusable functions, improving maintainability and reducing rerun errors across experiments.

PROJECTS & ACTIVITIES

Cornell Break Through Tech Fellowship
Machine Learning and AI Fellow *May 2026 – Present*

- Selected for a competitive year-long AI/ML fellowship; completed a technical 12-week AI Program from Cornell Tech, building and validating ML models on real-world datasets in Python.
- Completing a technical, industry-relevant project, and virtual professional development and technical workshops on topics including authentic leadership, effective collaboration, project management, and professional norms.

Gamma-Scalping Optimal-Exit Research Backtester *April 2026 – July 2026*
https://github.com/timmyhsu07/gamma-exit.git

- Built a research-grade options backtester (Python, NumPy, SciPy, pandas) implementing a *published* exit-timing framework (*Ramkumar, 2025*), validating Black–Scholes/Greeks against QuantLib to **1e-8** across **3,000+ Monte Carlo paths.**
- Identified a drift-term inconsistency in the paper's hedged-P&L derivation via invariance testing, then quantified its "optimal-exit" benchmark as a non-tradable foresight ceiling (**+55%** mean P&L vs. hold-to-expiry, net **+66%** after transaction costs).
- Designed a look-ahead-proof backtest architecture with immutable per-day state and a quarantined foresight benchmark; showed a causal vol-forecast exit rule captures **72%** of that ceiling's edge (**90% CI: 66–77%**) net of costs, vs. ~0% with no detectable edge (backed by a **172-test suite** and a write-once data cache).

Quant Challenge 2025 Competition – Top 2% Finalist *Sept 2025*
https://github.com/timmyhsu07/QuantChallenge25-Test-Repo.git

- Engineered an end-to-end machine learning pipeline using Random Forests (200 trees, max depth = 30) to analyze **10,000+ observations** across **14 engineered features**, improving predictive accuracy by **>25%** over baseline models.
- Implemented feature selection, correlation analysis, ensemble modeling, and probability calibration to improve model generalization and robustness.
- Developed Courtsense, a live basketball trading algorithm processing **1,000+** in-game events per match, optimizing bet sizing via the Kelly Criterion and achieving a simulated **18% ROI** across test scenarios, validated through backtesting and stress testing across varying game states and volatility conditions.

SKILLS

Languages: Python, Java, C++, SQL,
ML/Data Science: PyTorch, scikit-learn, NumPy, Pandas, Random Forests, dimensionality reduction (PCA, t-SNE, UMAP, PaCMAP), autoencoders, feature engineering, model validation
Tools & Platforms: Git, GitHub, OpenCV, scikit-image, Matplotlib, Jupyter, LaTeX, VS Code, Docker, DuckDB, Parquet, QuantLib, pytest,
Spoken Languages: Chinese (Native), English (Native)